Portland State
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Excel Best Practices For Students and Early Professionals

Designed for new and growing users, this guide helps you build clean, readable spreadsheets for class, work, or personal use.

Plan Before You Build

- **Sketch it out first,** even if it's on the back of a paper napkin. Outline what data you'll include, what calculations are needed, and what the final result should show. A rough drawing helps organize your thinking before jumping into Excel.
- Think in terms of **inputs and outputs**. What data will go into the spreadsheet, and what do you want to learn or show from it? Start with that big picture before entering any values. *(Tip: Your layout will depend on whether your list grows down the sheet or across. See "How the Layout Grows" near the end of this guide.)*

Make It Easy to Read at a Glance

- Remove unnecessary decimal places to reduce visual clutter and make scanning easier.
- Use borders, bold headers, and light shading to help important information stand out.
- Freeze panes so column headers stay visible when scrolling through data.
- Use standard, readable fonts like Calibri or Arial, and avoid novelty fonts like Comic Sans that can look unprofessional.
- Apply comma formatting to large numbers, even if they aren't currency, to improve clarity and speed up interpretation.
- Avoid random empty rows and columns. These gaps may seem harmless, but they disrupt the flow of data and can interfere with sorting, filtering, and formulas.
- **Design matters.** Use colors with care and stick to muted tones or soft highlights instead of neon or bright red, which can be hard on the eyes or signal errors. Always maintain high contrast between text and fill color. If the background is dark, use white text and consider bolding it for added readability.

Structure Data Clearly

- Keep **one piece of data per cell**, such as separating first names and last names or dates and values.
- **Label** each column clearly using headers, and add descriptive text near important numbers or calculations, so it's easy to understand what each number represents.

Ensure Your Data Is Fully Visible

- **Adjust column widths** manually so that all content is fully visible.
- Seeing `#######` instead of numbers? That just means the column is too narrow. Widen it to display the full value.
- Use **AutoFit** (double-click the right edge of the column header) for a quick and clean adjustment.
- Use **Wrap Text** when column content is too long, but make sure the row height adjusts so all the text remains visible and easy to read.

Use Formulas the Right Way

- **Always reference cells** instead of typing numbers directly into formulas.
- Avoid **hard coding**, which means typing static values directly into formulas. This makes updates harder and increases the chance of errors.
- Place formulas to the right of, or below the data, for logical flow.
- Use `SUM()` instead of adding cells one by one: `=SUM(A1:A4)` is cleaner and easier to manage than `=A1+A2+A3+A4.`
- Use parentheses to control the order of calculation. Excel follows math rules, so `(A1+B1)*C1` gives a different result than `A1+B1*C1`.

Check Your Work

- **Check** formulas after copying, editing, or moving data.
- Use Excel's **formula preview** colors to quickly see which cells are referenced.
- Compare values using **AutoCalculate,** in the status bar, to confirm accuracy.
- **Double-check spelling** in headers and labels. Clean labels make your spreadsheet easier to read and more professional, Excel doesn't autocorrect like Word does.

Sort and Organize for Better Insights

- **Sort** data when it helps tell a story or improves usability, such as alphabetically, by date, or by value.
- **Group** related content together, and follow a consistent layout.

Use Efficiency Tools

- **Efficiency** isn't just a bonus, it's a core workplace skill. Learning shortcuts saves time, reduces errors, and shows attention to detail.
- **Use AutoFill** to quickly copy formulas, extend sequences (dates, months, numbered lists), or apply formatting across rows and columns.
- **Use the Format Painter tool** to quickly copy formatting from one cell to others. This helps keep your spreadsheet consistent without manually reapplying styles, borders, or number formats over and over.
- **Use Undo (Ctrl+Z or Command-Z).** If you make a mistake, undo it instantly. It encourages learning and reduces hesitation.
- **Use keyboard shortcuts.** Even learning basic shortcuts like Ctrl+C (copy), Ctrl+V (paste), or Ctrl+Arrow Keys (to navigate quickly) can save time and improve workflow.

Choose the Right Layout: Vertical vs. Horizontal

Common Layout for Class and the Workplace

- Most spreadsheets in class use a horizontal layout, with categories in the top row and data flowing across rows. This makes it easier to apply functions like SUM and AVERAGE, especially when comparing data across time (tracking categories across months).

When To Use a Vertical Layout

- Use a vertical layout when you're building a list that grows row by row, like expense tracking, planning a budget, or organizing tasks. Each new item is added as a new row, while columns stay fixed for things like budget item, quantity, and total cost.
- Some vertical spreadsheets follow a **fixed** structure that's set up in advance, like loan payment tables or calendars. These might not grow over time, but they still organize information top to bottom, one item per row.

How the Layout Grows

- In a vertical layout, new data is added in new rows, the sheet grows downward, with each entry added at the bottom. In a horizontal layout, new data is added in new columns, the sheet grows to the right
- Whichever you choose, keep the layout consistent so others can follow your structure easily.